

Ana Alice Pimenta Pereira

ap2375@cornell.edu · anapimentapereira.com · linkedin.com/in/anapimentapereira · Ithaca, NY 14850

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EDUCATION

Cornell University

Ithaca, NY

Ph.D., Biomedical & Biological Sciences · August 2024 – Expected 2029

GPA: 4.00 / 4.00

Dissertation (working title): “Alphavirus–host arms race on RNA recognition and stress responses”

Advisor: Dr. Rafael Kroon Campos

Federal University of Viçosa (UFV)

Viçosa, Brazil

D.V.M., Doctor of Veterinary Medicine · March 2019 – July 2024

Ranked 2nd of 60 students · GPA: 87.5 / 100

Degree requirement: supervised internships at a federal reference laboratory of the Brazilian Ministry of Agriculture, Livestock and Supply (MAPA), in the Viral Disease Diagnosis Laboratory (LDDV) and the Bacterial Disease Diagnosis Laboratory (DDB).

RESEARCH EXPERIENCE

Graduate Researcher, Virology

2025 – Present

Dr. Rafael Kroon Campos · Kroon Campos Lab, Cornell University, Ithaca, NY

Areas of investigation: alphavirus biology, RNA modifications (m6A, m5C), innate immunity and stress responses, stress granules, Toll signaling, mosquito vector competence

Project: I am defining how RNA modifications, such as N6-methyladenosine (m6A) and 5-methylcytosine (m5C), remodel antiviral gene expression and stress signaling during alphavirus infection. I am determining how m6A regulates canonical stress granules and the virus-induced condensates formed by nonstructural protein 3 (nsP3). I am dissecting how nsP3 interactions with the Toll ligand Spätzle and with phenoloxidase reprogram mosquito immunity and facilitate viral dissemination and transmission.

Rotation Student

Oct – Dec 2024

Dr. Deborah Fowell · Fowell Lab, Cornell University, Ithaca, NY

Areas of investigation: immunology, germinal center biology, B cell differentiation, T follicular helper cells, transgenic mouse models

Project: The goal of this rotation was to determine the temporal requirements for T follicular helper (Tfh) cell support of germinal center (GC) responses. I tracked memory B cells (Bmem), plasma cells, and long-lived plasma cells (LLPCs), and I used the S1PR2-iCRE x stop-floxed-tdTomato transgenic mouse model, which fluorescently labels germinal center derived cells, to follow their differentiation at defined temporal checkpoints.

Rotation Student

Aug – Oct 2024

Dr. Sarah Caddy · Caddy Lab, Cornell University, Ithaca, NY

Areas of investigation: viral immunology, antibody effector functions, canine distemper virus, natural killer cell biology

Project: The goal of this rotation was to characterize immune responses to canine distemper virus (CDV), with a focus on antibody-dependent cellular cytotoxicity (ADCC), and to identify immunological gaps relevant to canine health. I worked on developing and optimizing a protocol for purifying natural killer (NK) cells from canine whole blood, which is a prerequisite for assessing the functionality of these cells.

Leadership Program for Veterinary Students

May – Aug 2023

Dr. Sarah Caddy · Caddy Lab, Cornell University, Ithaca, NY

Areas of investigation: viral immunology, antibody-mediated phagocytosis, canine parvovirus, macrophage biology

Project: The goal of this program was to study the antibody response to canine parvovirus (CPV). I found evidence of both phagocytosis and antibody-mediated phagocytosis of the virus by canine macrophages, and I presented this work as first author at two conferences. I joined the program as the first Brazilian selected in its thirty-year history, and I received financial support from the Cornell Richard P. Riney Canine Health Center.

Undergraduate Researcher, PIBIC/CNPq

2023

Dr. Alex Pauvalid-Corrêa · Viçosa Veterinary Virology Lab (LAVEV), UFV, Viçosa, Brazil

English: PIBIC/CNPq = Institutional Program for Scientific Initiation Scholarships, funded by Brazil's National Council for Scientific and Technological Development.

Areas of investigation: arbovirus serosurveillance, flaviviruses, equine virology, One Health, science outreach

Project: I assessed the presence of neutralizing antibodies to Ilheus virus (ILHV) in equid populations in Viçosa, Brazil, using the plaque reduction neutralization test (PRNT), and I standardized the same test for West Nile virus (WNV). I coordinated undergraduate team members, delegated tasks, managed schedules, and oversaw laboratory organization. I engaged the rural population of Viçosa on active viral surveillance, which contributed to bridging the gap between academia and society.

Undergraduate Researcher, PIBIC/CNPq

2021 – 2022

Dr. Abelardo Silva-Júnior · Animal Virology Laboratory (LIVA), UFV, Viçosa, Brazil

Areas of investigation: animal virology, recombinant protein expression, Senecavirus A

Project: I expressed recombinant Senecavirus A (SVA) genes in *Escherichia coli* in order to produce viral antigens for downstream serological applications. I held a National Council for Scientific and Technological Development (CNPq) Level A scholarship.

Undergraduate Researcher, voluntary PIVIC/CNPq

2020 – 2021

Dr. Abelardo Silva-Júnior · Animal Virology Laboratory (LIVA), UFV, Viçosa, Brazil

English: PIVIC = the volunteer (unpaid) track of the same national program.

Areas of investigation: swine virology, molecular epidemiology, SARS-CoV-2 diagnostics, canine parvovirus

Project: I contributed to a study of Porcine Circovirus 3 (PCV3) in swine farms that identified correlations with reproductive failures. During the first year of the coronavirus disease 2019 (COVID-19) pandemic I quantified severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) viral load in samples from Minas Gerais, as first author, and I contributed to an epidemiological analysis of viral circulation across the Manhuaçu, Ponte Nova, and Ubá regional health units. I also contributed to a comparison of Brazilian commercial vaccine strains with circulating field strains of canine parvovirus type 2 (CPV-2) at the antigenic VP2 protein.

High School Research Investigation

2016 – 2018

SESI CAT Maria José D'Almeida Mello, Brazil

Areas of investigation: photocatalysis, materials chemistry, environmental science

Project: I designed and developed a photocatalytic paint based on titanium dioxide (TiO₂) and lime for the conversion of pollutant gases. I created experimental protocols using accessible materials, including polyethylene terephthalate (PET) bottles, lime, and water, in order to evaluate its effectiveness. I won awards at national and international science fairs, and I produced the first publication of my career.

PUBLICATIONS – ACADEMIC

Published Manuscripts

1. **Pimenta-Pereira AA**, Lubinski B, Campos RK. (2026) Alphavirus nsP3 as a multifunctional orchestrator of virus–host interplay. *Journal of Virology*, 100(7), e01779-25. doi.org/10.1128/jvi.01779-25 **[Review]**
2. Teh S, Truyen LH, Woodyear S, **Pimenta-Pereira AA**, Peprah AAE, Lanman E, Lonergan TM, Reid A, Cheong SH, Caddy SL. (2026) Quantifying maternal antibody transfer to colostrum and cord blood reveals virus-specific selectivity in dogs. *Frontiers in Immunology*, 16, 1753521. doi.org/10.3389/fimmu.2025.1753521
3. Pauvalid-Corrêa A, de Jesus LBG, Barbosa VAP, Oliveira YET, da Silva MAN, de Paula Rebello M, de Freitas GS, Bem GS, **Pimenta-Pereira AA**, Rabelo AGC, Gonçalves EVJD, Moreira MF, Morente BB, Tomich MS, Domingues LSS, Magalhães MML, Cavalheiro TCBS, do Lago EP, de Barros JJF, Barreto-Vieira DF. (2025) Isolation of Orthopoxvirus vaccinia during an outbreak of vesicular disease in dairy cattle from Viçosa, state of Minas Gerais,

southeast Brazil. *Brazilian Journal of Microbiology*, 56(4):2619–2625. doi.org/10.1007/s42770-025-01719-5. PMID: 40745132

4. **Pereira AAP**, Duarte AA, Ramos APC. (2017) Produção de tinta à base de dióxido de titânio capaz de transformar gases poluentes em gases inofensivos no município de Pedro Leopoldo. *Faculdade ASA*, v. I, n. 26, Brumadinho, p. 159–172.

English: “Production of titanium dioxide-based paint capable of converting pollutant gases into harmless gases in the municipality of Pedro Leopoldo.” [view PDF](#)

INVITED TALKS

2025 | International Veterinary Students' Association (IVSA)

“Carreira Internacional na Medicina Veterinária”

English: “International careers in veterinary medicine.”

2024 | Universidade Estadual do Maranhão (UEMA)

Maranhão State University, Brazil

“Oportunidades de estágio para alunos de medicina veterinária no exterior”

English: “Internship opportunities abroad for veterinary medicine students.”

SELECTED POSTERS AND PRESENTATIONS

Sep 2026 | M&I Trainee Rapid-Fire Talk, Dept. of Microbiology & Immunology Retreat

Cornell University

Pimenta-Pereira AA. “Through the RNA Looking-Glass.” [\[oral presentation · award winner\]](#)

Sep 2024 | Simpósio de Pesquisa em Medicina Veterinária

UFV, Viçosa, Brazil

Bem GS, de Jesus LBG, de Freitas GS, **Pimenta-Pereira AA**, Lima RAL, Pauvolid-Corrêa A. “Investigação de anticorpos para o vírus Ilhéus em equídeos da região de Viçosa, Minas Gerais.”

English: Investigation of antibodies to Ilhéus virus in equids from the Viçosa region, Minas Gerais. doi.org/10.13140/RG.2.2.27610.96960

Nov 2023 | VI International Symposium of Microbiology and Biotechnology

de Jesus LBG, **Pimenta-Pereira AA**, de Freitas GS, Barbosa VAP, do Lago EP, Pauvolid-Corrêa A. “Exploring a vesicular and pustular disease outbreak in dairy cattle and milkers in the microregion of Viçosa, state of Minas Gerais, Brazil.”

Oct 2023 | Simpósio de Integração Acadêmica

UFV, Viçosa, Brazil

Pimenta-Pereira AA, Brito B, de Jesus LBG, Lima RAL, Ferraz CM, Pauvolid-Corrêa A. “Padronização do teste de neutralização por redução de placa para o vírus do Oeste do Nilo.” [\[first author\]](#)

English: Standardization of the plaque reduction neutralization test (PRNT) for West Nile virus.

Oct 2023 | Simpósio de Integração Acadêmica

UFV, Viçosa, Brazil

de Jesus LBG, **Pimenta-Pereira AA**, Mascarenhas IN, Bem GS, ... Pauvolid-Corrêa A. “Ausência de evidência de exposição ao vírus Mayaro em saguis de vida livre de área antropizada da Zona da Mata de Minas Gerais.” [\[oral presentation\]](#)

English: No evidence of Mayaro virus exposure in free-ranging marmosets from an anthropized area of the Zona da Mata, Minas Gerais.

Sep 2023 | 34th Brazilian Congress of Virology

Ouro Preto, Brazil

Pimenta-Pereira AA, Woodyear SAK, Chandler TL, Weichert WS, Caddy S. “Production of canine parvovirus virus-like particles to study Fc-mediated effector functions of canine antibodies.” [\[first author\]](#)

Sep 2023 | 34th Brazilian Congress of Virology

Ouro Preto, Brazil

Jesus LBG, Morente BB, Souza IF, Oliveira GAS, Kaku GY, Bem GS, **Pereira AAP**, Freitas GS, Ferraz CM, Lima RAL, Moreira MF, Veloso ACV, Costa YV, Gonçalves EVJD, Teixeira CS, Silva YFRS, Pauvolid-Corrêa A. “Investigation of West Nile virus exposure in equids from Viçosa, state of Minas Gerais, Southeast Brazil.”

2023 | Veterinary Scholar Symposium

San Juan, Puerto Rico

Pimenta-Pereira AA, Woodyear SAK, Chandler TL, Weichert WS, Caddy S. “Developing a canine antibody phagocytosis assay to study maternal antibodies specific for canine parvovirus.” **[first author]**

Jul 2023 | V FAMERP–UTMB: Emerging Infections in the Americas *São José do Rio Preto, SP, Brazil*

de Jesus LBG, Brito B, Lima RA, Ferraz CM, **Pimenta-Pereira AA**, Oliveira GAS, Freitas M, Kaku GY, Souza IF, Bem GS, Veloso ACV, Mascarenhas IN, Marques AM, de Melo FR, Voorwald F, Pauvolid-Corrêa A. “Lack of evidence of Mayaro virus exposure in free-ranging marmosets of an anthropized area of Zona da Mata, Southeast Brazil.”

2022 | 26th International Pig Veterinary Society Congress (IPVS) *Rio de Janeiro, Brazil*

Toledo LT, Lino LF, Gonzaga NF, Marks FS, Pereira CER, **Pereira AAP**, Diamantino CA, Oliveira LG, Silva-Júnior A. “Virulence differences between *Mycoplasma hyopneumoniae* Brazilian strains at experimentally infected swines.” *Proceedings IPVS 2022, p. 366. Presented by Leonardo Toledo.*

2020 | XXXI Brazilian Congress of Virology & XV Mercosur Virology Meeting *online, Porto Alegre, Brazil*

Pereira AAP, Leber LLP, Santos MR, Barros RA, Zerbini PA, Zerbini Junior FM, Silva LMN, Assao VS, Silva Junior A. “Semiquantitative evaluation of viral load of SARS-CoV-2 in samples from Minas Gerais, Brazil.” **[first author]**

2020 | XXXI Brazilian Congress of Virology & XV Mercosur Virology Meeting *online, Porto Alegre, Brazil*

Silva LMN, Santos MR, Assao VS, Carvalho JA, Leber LLP, Rosado NCL, Gomes LL, Santos JC, **Pereira AAP**, Hubner SO, Silva Junior A. “Brazilian commercial vaccines against CPV-2 genetically differ from field strains in the antigenic protein VP2.” **[panelist]**

AWARDS AND RECOGNITIONS

Sep 2026 | M&I Trainee Rapid-Fire Talk Winner *Cornell University*

Department of Microbiology & Immunology Retreat Award Series. Awarded for winning the M&I Trainee Rapid-Fire Talk and “demonstrating excellence in science communication” (9 September 2026).

2025 | Líder Estudiar Finalist *“the Brazilian Rhodes Scholarship”*

Selected as a finalist (0.5% acceptance to this stage) for the 2025 Líderes Estudiar Leaders Scholarship Program — the most competitive scholarship in the country.

2023 | Best Poster Presentation Award *34th Brazilian Congress of Virology*

Awarded in the “Veterinary Virology: Undergraduate” category for “Production of Canine Parvovirus Virus-Like Particles to Study Fc-Mediated Effector Functions of Canine Antibodies.”

2023 | Cornell Richard P. Riney Canine Health Center Award

Received financial support for summer studies at Cornell University focused on advancing research in canine health.

2023 | PIBIC/CNPq Research Funding

Awarded by the Brazilian government to conduct research on equine serosurveillance.

2021 | PIBIC/CNPq Research Funding

Awarded by the Brazilian government to carry out research in animal virology.

2017 | Technological Innovation Award

USP Innovation Agency

Awarded best project in technological innovation at the Brazilian Science and Engineering Fair (FEBRACE).

2017 | Most Outstanding Exhibit in Materials Science

ASM Materials Education Foundation

Awarded for excellence in materials science at the Brazilian Science and Engineering Fair (FEBRACE).

2017 | 2nd Place, Exact and Earth Sciences

FEBRACE

Awarded at the Brazilian Science and Engineering Fair for outstanding research in Exact and Earth Sciences.

SCIENCE COMMUNICATION AND TEACHING

Teaching

Teaching Assistant

Fall 2025

Teaching Assistant

2022 – 2023

VET 140 — *Veterinary Microbiology, Universidade Federal de Viçosa*

- Led the laboratory classes for the course.

Teaching Assistant

2023

VET 340 — *Bacterial Diseases, Universidade Federal de Viçosa*

Teaching Assistant

2022

VET 331 — *Veterinary Anesthesiology, Universidade Federal de Viçosa*

Science Communication

Winner, M&I Trainee Rapid-Fire Talk

Sep 2026

Department of Microbiology & Immunology Retreat, Cornell University

- Competitive short-format research talk judged across departmental trainees; recognised by the department for excellence in science communication.

Science Communication Intern

2023

CPT: VET Profissional, Brazil

- Created a format for communicating veterinary concepts on social media, and launched the company's TikTok channel.
- My first video reached more than 500,000 views. The channel has since grown to 20,400 followers, alongside 172,000 followers on Instagram.
- Produced the underlying content: image and video editing, posts, and articles.

OUTREACH & SERVICE

Member, BBS Graduate Student Society (GSS)

2026 – Present

Biomedical & Biological Sciences PhD Program, Cornell University

BBS Peer Mentor

Aug 2026 – Present

Biomedical & Biological Sciences PhD Program, Cornell University

- Supporting incoming and current students in the program.

Organizer

Aug 2026

25th Cornell Biomedical & Biological Sciences Symposium

Event Organizer, Social Impact Researchers Award

2025 – 2026

Brazil Conference at Harvard & MIT

- Organized an award created to recognize research of excellence with the potential to transform science and the reality of Brazil, and to encourage science communication.
- The Brazil Conference at Harvard and MIT is the largest annual conference on Brazil, convening a global audience to discuss the country's future and its role in the world.

Marketing Director

2020

XXII Semana Acadêmica de Medicina Veterinária (SAMEV), UFV

English: XXII Academic Week of Veterinary Medicine.

- Managed the marketing schedule, promoted the event, and addressed inquiries from the public via email and social media platforms.

President

2019

Wildlife Study Group, UFV

- Facilitated a team of 12 individuals in organizing a study group focused on the care of wild animals, addressing a gap in the curriculum at the time, as the university had not yet offered a course on the subject.

Public Affairs Director

2019

Wildlife Study Group, UFV

- Organized, received, and presented speakers and other guests for weekly meetings.

OTHER WORK EXPERIENCE

Diagnostics Intern, Animal Virology and Bacteriology

Jan – Feb 2023

Federal Laboratory for Agricultural and Livestock Defense (LFDA), Ministry of Agriculture, Livestock and Supply (MAPA), Pedro Leopoldo, Minas Gerais, Brazil

- Followed diagnostic workflows in facilities operating at biosafety levels NB2, NB3, and NB4.

D.V.M. Intern

2021

Atlas Hospital Veterinário (Atlas Veterinary Hospital), Brazil

- Intern in a small animal hospital setting with a focus on internal medicine, hospitalization, surgery, and small animal anesthesia.
- Handled medication for hospitalized animals and assisted in surgical procedures and patient care.
- Participated in company-offered courses, presented seminars, and engaged in clinical case discussions related to dog and cat care.

Extension Project — Família do Leite (Milk Family)

2021

Universidade Federal de Viçosa

- Herd management: oversaw the general management of the herd, including calf management, medication administration, and vaccination.
- Routine observation: observed the daily operations of various sectors related to animal health, reproduction, calf rearing, and nutrition.
- Technical education: participated in directed studies and attended technical lectures on dairy herd handling, feeding, and health.

IN THE NEWS

2017 | FEBRACE / LSI-TEC, Universidade de São Paulo

Brazilian Science and Engineering Fair

“Produção de tinta à base de Dióxido de Titânio e Cal capaz de transformar poluentes atmosféricos” — project presentation, [watch on YouTube](#)

English: “Production of titanium dioxide and lime-based paint capable of transforming atmospheric pollutants.” FEBRACE is Brazil’s national science and engineering fair.

2017 | Futura Profissão, SESI

interview · television series

“O ferro, o fogo e o filtro,” Episode 06 — [watch on YouTube](#)

English: “Future Profession — Iron, fire, and the filter,” a SESI television series on science and careers.